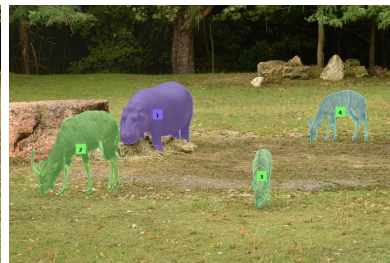
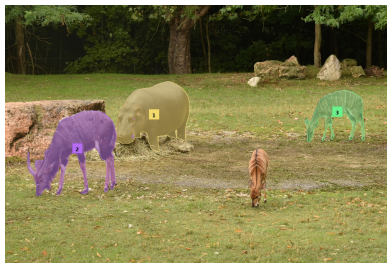


Input Image

Per-Token Format

SetCon (ours)



Missing masks

Complete masks

"If a zookeeper wants to identify the main herbivores currently feeding in this scene, which objects should be detected?"

(a) Comparison in handling open-ended referring scenarios

Input Image

Per-Token Format

SetCon (ours)



Duplicate masks

Distinct masks

"If someone wanted to pull down all the curtains in the room, which ones would they have to address and can you describe them?"

Open-Ended Referring Query

USER: If you wanted to make a refreshing summer vegetable salad, which objects in the image would be the best to pick first?

Previous Per-Token Format

Duplicated or missing masks

ASSISTANT: You should first pick the fresh, green cucumber [SEG] and then select the ripe, red tomatoes [SEG] [SEG] [SEG] [SEG] [SEG] [SEG] .

Flat sequence of indistinguishable [SEG] tokens

SetCon (ours)

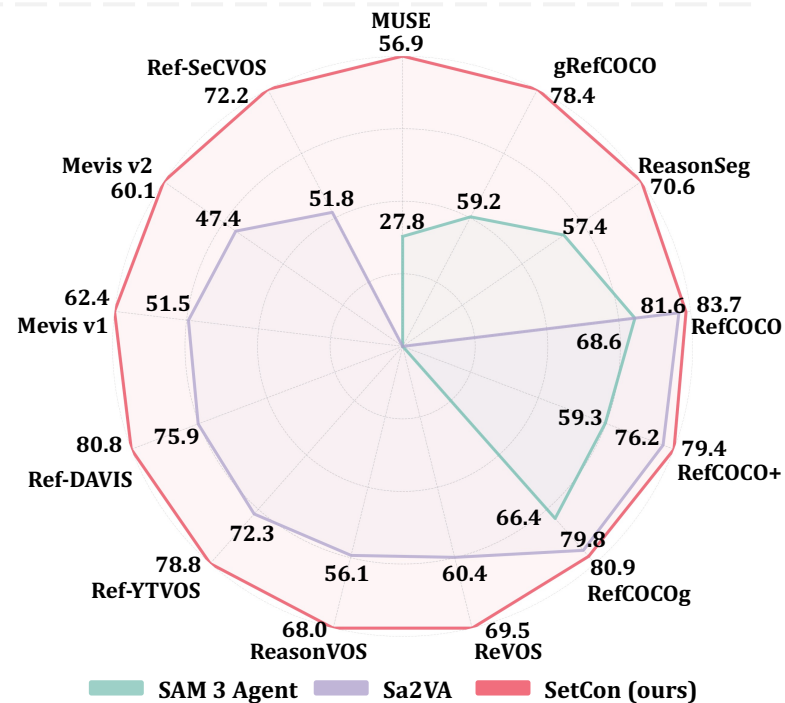
More distinct and complete mask sets

ASSISTANT: It's **summer salad staples**. (includes **green cucumber** and **red tomato**).

Set-level prediction via hierarchical language-grounded concepts



(b) Schematic comparison between our SetCon and previous format



(c) Quantitative results